

Group Discussion 3

RNA-seq Workflow Design Assignment

Choose a disease of interest, or use the disease assigned to you, and design a complete **RNA-seq and differential expression analysis workflow** using **Nextflow** or another workflow system of your choice. Your goal is to outline a reproducible transcriptomics pipeline from raw sequencing reads to biological interpretation.

Your submission should include the following:

1. **Pipeline overview**
Briefly describe the major steps in your workflow, beginning with raw FASTQ files and ending with differential expression and pathway analysis.
2. **Shell script or workflow code**
Provide a shell script, Nextflow script, or equivalent code showing the specific commands and parameter settings you would use.
You may use the same tools described in the original paper if the authors reported their sequencing and preprocessing methods.
3. **Justification of parameter choices**
Explain the rationale for the key parameter settings you selected for each tool used in sequence processing.
You do not need to explain file paths or directory structure.
4. **Reference files and databases**
State which reference genome, gene annotation, or gene set resources you would use, and identify the database or source for each one (for example, Ensembl, GENCODE, NCBI, MSigDB, GO, KEGG, Reactome).
5. **RNA-seq processing choices**
Explain your choice of tools for:
 - read quality control and trimming
 - alignment or pseudoalignment
 - read quantification
6. **Differential expression and enrichment analysis**
Explain your choices for the parameters used in:
 - **DESeq2**
 - **GSEA**
 - **gprofiler2**

Focus especially on any settings that differ from the defaults.
7. **Biological interpretation**
Describe the biological questions that can be addressed by the contrasts you set up in your differential expression analysis.

8. **Limitations and sources of bias**

Discuss important limitations, assumptions, and possible biases in the workflow you designed.

Formatting requirements

- Limit the **main narrative** to **one single-spaced page**
- Use **12-point Calibri, Arial, or Aptos**
- You may place **code** and **references** on separate page(s)

Sample

Disease: Colorectal cancer (CRC)

Biological question: Which genes and pathways are differentially expressed in colorectal tumor tissue compared with adjacent normal colon tissue?

I would design a bulk RNA-seq workflow in Nextflow beginning with paired-end FASTQ files, followed by read quality control, adapter/quality trimming, genome alignment, gene-level quantification, differential expression testing, and pathway enrichment. I chose this design because it is reproducible, modular, and appropriate for comparing transcriptomic differences between CRC tumor and matched or unmatched normal tissue.

For preprocessing, I would first evaluate raw read quality with **FastQC** and summarize reports with **MultiQC**. I would trim reads using **Trim Galore** with default adapter detection plus a minimum read length filter (for example `--length 20`) to remove very short reads that may align nonspecifically. I would keep trimming conservative to avoid unnecessary loss of informative sequence. For alignment, I would use **STAR** against the **GRCh38/hg38** human reference genome with a matching **GENCODE** gene annotation GTF. STAR is appropriate because it is splice-aware, fast, and widely used for mammalian RNA-seq. I would use parameters such as `--outFilterMultimapNmax 10` to limit excessive multimapping and `--quantMode GeneCounts` to generate gene-level counts directly from aligned reads. If I wanted transcript-level quantification instead, I could use **Salmon** against a GENCODE transcriptome index and summarize to genes with `tximport`.

For differential expression, I would use **DESeq2** with a design formula such as `~ batch + condition`, where condition is tumor versus normal. Including batch is important if libraries were prepared or sequenced across different runs. I would prefilter very low-count genes to reduce noise, for example retaining genes with at least 10 total counts across samples. I would use the Wald test for the main tumor-versus-normal contrast and adjust p-values with Benjamini-Hochberg FDR. If effect size stabilization is needed for ranking or visualization, I would apply `lfcShrink`.

For pathway-level interpretation, I would run **GSEA** using a ranked gene list based on shrunken log2 fold change or signed statistical values. I would use **MSigDB Hallmark** gene sets because they reduce redundancy and capture broad biological programs such as inflammatory signaling, MYC targets, and epithelial–mesenchymal transition. I would also use **gprofiler2** for over-representation analysis on significant DE genes and report GO Biological Process, Reactome, and KEGG terms. I would set the organism to human and apply multiple-testing correction; any nondefault choices, such as restricting gene set size or focusing only on annotated genes, would be justified to improve interpretability.

The main contrast, tumor versus normal, answers which genes are upregulated or downregulated in CRC and which pathways are most perturbed. Additional contrasts could compare stage groups, MSI versus MSS tumors, or treatment responders versus nonresponders if metadata are available. Limitations include batch effects, tumor heterogeneity, variable RNA quality, incomplete clinical covariates, and biases introduced by trimming, alignment ambiguity, or annotation choice. Results are also sensitive to sample size and the accuracy of metadata.

```
#!/bin/bash
```

```
# Example bulk RNA-seq workflow for colorectal cancer
```

```
# Paired-end reads: sample_R1.fastq.gz and sample_R2.fastq.gz
```

```
# 1. Quality control
```

```
fastqc sample_R1.fastq.gz sample_R2.fastq.gz -o qc_raw/
```

```
# 2. Trimming
```

```
trim_galore \
```

```
--paired \
```

```
--quality 20 \
```

```
--length 20 \
```

```
--fastqc \
```

```
-o trimmed/ \
```

```
sample_R1.fastq.gz sample_R2.fastq.gz
```

3. Alignment with STAR

```
STAR \
```

```
--runThreadN 8 \
```

```
--genomeDir /ref/STAR_GRCh38_GENCODE/ \
```

```
--readFilesIn trimmed/sample_R1_val_1.fq.gz trimmed/sample_R2_val_2.fq.gz \
```

```
--readFilesCommand zcat \
```

```
--outFileNamePrefix aligned/sample_ \
```

```
--outSAMtype BAM SortedByCoordinate \
```

```
--outFilterMultimapNmax 10 \
```

```
--quantMode GeneCounts
```

4. Summarize QC

```
multiqc . -o multiqc_report/
```

Example DESeq2 / GSEA / gprofiler2

```
dds <- DESeqDataSetFromMatrix(
```

```
  countData = counts,
```

```
  colData = metadata,
```

```
  design = ~ batch + condition
```

```
)
```

```
dds <- dds[rowSums(counts(dds)) >= 10, ]
```

```
dds <- DESeq(dds)
```

```
res <- results(dds, contrast = c("condition", "Tumor", "Normal"))
```

```
res_shrunk <- lfcShrink(dds, coef="condition_Tumor_vs_Normal", type="apeglm")
```

GSEA

```
library(fgsea)
```

```
ranks <- res_shrunk$log2FoldChange
```

```
names(ranks) <- rownames(res_shrunk)
```

```
ranks <- sort(na.omit(ranks), decreasing = TRUE)
```

gprofiler2

```
library(gprofiler2)
```

```
sig_genes <- rownames(subset(res, padj < 0.05 & abs(log2FoldChange) > 1))
```

```
gost_res <- gost(
```

```
  query = sig_genes,
```

```
  organism = "hsapiens",
```

```
  sources = c("GO:BP", "REAC", "KEGG"),
```

```
  correction_method = "fdr"
```

```
)
```

Biological questions answered by the contrast

With the contrast **Tumor vs Normal**, students can answer:

- Which genes are upregulated in colorectal tumors?
- Which genes are downregulated in tumors relative to normal tissue?
- Which biological pathways are enriched among tumor-associated genes?
- Do tumor samples show signatures of proliferation, immune activation, metabolism changes, or epithelial–mesenchymal transition?

Limitations and biases to mention

A strong student answer should mention several of these:

- small sample size reduces power
- batch effects can confound disease effects
- poor RNA quality can distort counts
- trimming may remove useful sequence if too aggressive
- multimapping and annotation choice affect quantification
- tumor heterogeneity and stromal contamination may blur biological interpretation
- gene-level counting may miss transcript isoform changes
- pathway databases are incomplete and biased toward well-studied genes